

CHAPTER 3

Single-axis control

3.1. Space story

We started up a station-keeping maneuver on a Series 5000 spacecraft for the first time. Series 5000 were the latest GE momentum-bias spacecraft. Once the maneuver started, the performance was not what we expected so we shut it down. This was the first GE communications satellite with a Proportional-Integral-Derivative Controller. What we observed was due to a PID being conditionally stable and the minimum impulse being higher than expected. As it turned out, when you opened the valve just a little it burned more fuel getting more impulse per pulse. This was easy to fix. What was interesting was that the time response looked like a single-axis double integrator. This is the motivation for this chapter.

3.2. Introduction

This chapter presents a complete design of a single-axis control system including a Kalman filter for attitude determination. The goal is to present a simple system that illustrates many aspects of spacecraft control. It gives you code that you can explore to improve your understanding of spacecraft control. The overall system is shown in Fig. 3.1.

This controller is configured as a regulator. It maintains the angle but does not have a set point input. The set point is always zero.

3.3. Dynamical systems

The two dynamical systems that need to be modeled in this example are the spacecraft and the gyro. The spacecraft model is a double integrator

$$I\ddot{\theta} = T_c + T_d \quad (3.1)$$

or

$$\begin{aligned} \dot{\theta} &= \omega \\ I\dot{\omega} &= T_c + T_d \end{aligned} \quad (3.2)$$

where θ is the true angle and ω is the true angular rate, I is the inertia, T_c is the control torque, and T_d is the disturbance torque.

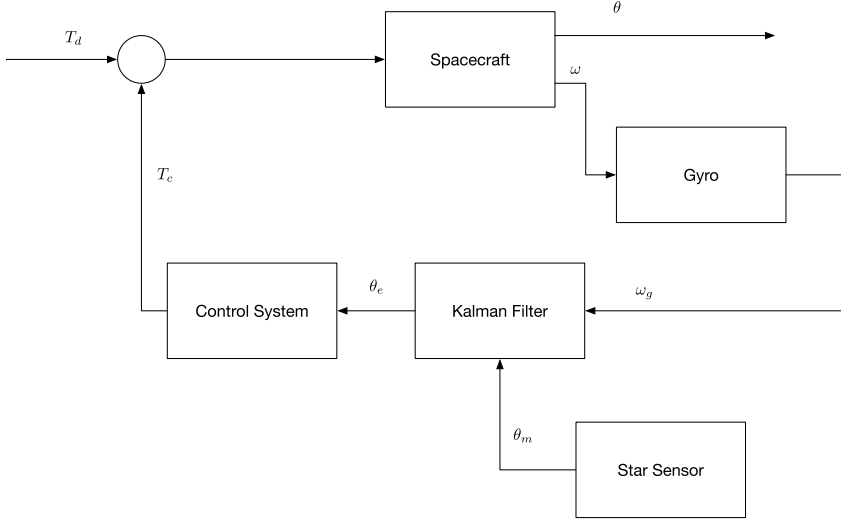


Figure 3.1 Single-axis control system.

The gyro model is a single integrator

$$\begin{aligned}\dot{\theta}_g &= \omega + b + v_\theta \\ \dot{b} &= v_b\end{aligned}\tag{3.3}$$

where θ_g is the integral of the true angular rate, ω plus the bias b . The bias is driven by white noise, meaning that it is a random-walk process. v_b is the bias noise input. v_θ is the angle white noise.

3.4. Control system

The control system is a proportional-derivative control system. In discrete time the controller is

$$T_c[k] = -K_f \left(\theta_e[k] + \tau_r \left(\frac{\theta_e[k] - \theta_e[k-1]}{\tau} \right) \right)\tag{3.4}$$

where K_f is the forward gain, τ_r is the rate damping time constant, τ is the controller time step, and k indicates the sample. If we write the continuous closed-loop system in the s -plane we get the second-order equation

$$Is^2 + K_f \tau_r s + K_f = 0\tag{3.5}$$

We can equate the gains with the undamped natural frequency and damping ratio

$$K_f = \omega_n^2 I\tag{3.6}$$

$$\tau_r = \frac{2\zeta\omega_n I}{K_f} \quad (3.7)$$

where ω_n is the desired undamped natural frequency and ζ is the damping ratio. If the sampling rate is sufficiently fast, these gains will give the desired response and we will not have to consider sampling time. Sufficiently fast is a sampling frequency much faster, that is five to ten times faster, than the control bandwidth, ω_n .

3.5. Kalman filter

The Kalman filter has two parts, state prediction and measurement incorporation. The gyro model in state-space form is

$$x_k = ax_{k-1} + b\omega_{k-1} \quad (3.8)$$

$$y = hx \quad (3.9)$$

where x is the Kalman filter state vector, h is the measurement matrix, and a is the state-transition matrix. It implements one time step for the state transitioning it from the state at $k-1$ to that at k . Expanding the matrices

$$\begin{bmatrix} \theta_g \\ b \end{bmatrix}_k = \begin{bmatrix} 0 & \Delta t \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \theta_g \\ b \end{bmatrix}_{k-1} + \begin{bmatrix} \Delta t \\ 0 \end{bmatrix} \omega_{k-1} \quad (3.10)$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \theta_g \\ b \end{bmatrix} \quad (3.11)$$

The state vector consists of the angle θ_g and the bias b . The equations show that the true angular rate and the bias both cause the state to change.

The Kalman-filter measurement incorporation step is

$$k = \frac{ph^T}{hph^T + r} \quad (3.12)$$

$$x_e = k(y - hx_e) \quad (3.13)$$

$$p = (E - kh)p \quad (3.14)$$

where E is a 2-by-2 identity matrix, p is the covariance matrix, r is the measurement covariance matrix, and the measurement matrix is h . The prediction step is

$$x_e[k] = ax_e[k] + b \left(\frac{\theta_i[k] - \theta_i[k-1]}{\Delta t} \right) \quad (3.15)$$

$$p = apa^T + q \quad (3.16)$$

where a is the state-transition matrix, q is the model-covariance matrix, and Δt is the time step. The model-covariance matrix is a Gaussian white-noise approximation to all

of the uncertainty in the model, including unmodeled inputs. The output of the rate-integrating gyro is differenced to get the rate. This is where the IMU connects to the model. The IMU output is not a measurement.

An important point is that the covariance follows a path predetermined by the measurement covariance, the model covariance, and the initial covariance. Essentially, it is a variable-gain filter that will reach steady-state values and stay there forever.

3.6. Simulation

Example 3.1 gives a simulation with the Kalman filter in the loop. The time step is chosen based on the bandwidth of the controller. It must be

$$\Delta T < \frac{1}{2} \frac{2\pi}{\omega_n} \quad (3.17)$$

This is known as the Nyquist sampling frequency. This can be understood by looking at a sine wave. Fig. 3.2 shows a sine wave. If we sampled at half the period, there is a possibility that we would see only zeros. Thus, we have to sample more frequently.

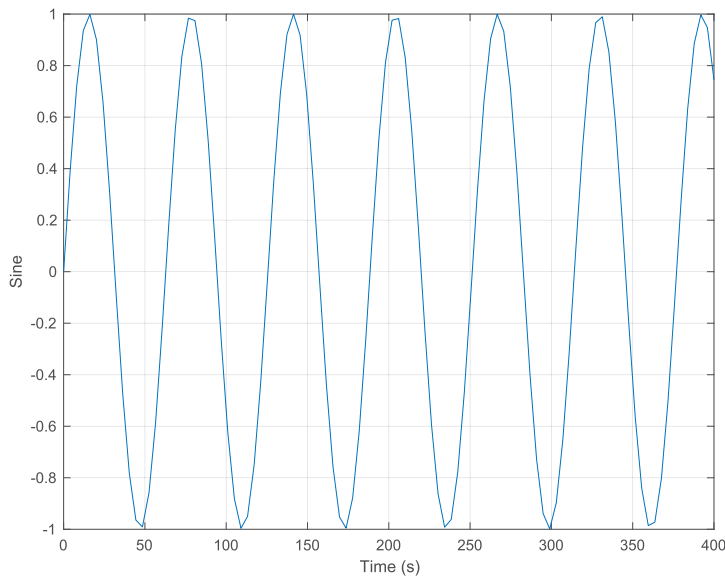


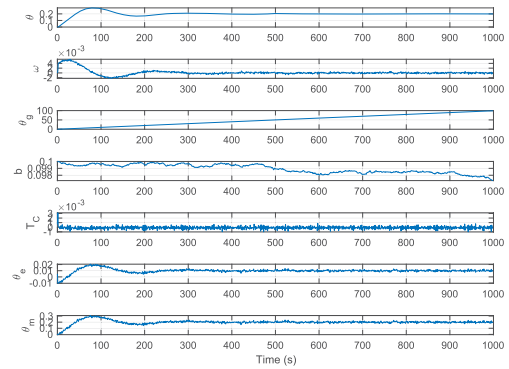
Figure 3.2 Sine wave to illustrate sampling. If sampling was exactly half the period, the samples might all be zero.

A step disturbance is applied to the spacecraft. The gyro measures the angular rate but its output includes the gyro bias, which is modeled with a random walk, and the angle noise. The integrated rate is used by the control system. Note that the integrated angle, from the gyro, grows with time due to the bias.

```

1 %% Simulation of a single axis spacecraft
2
3 %% Dynamical model data
4 tD = 1e-4; % Disturbance torque (Nm)
5 dRHS = struct('inertia',1,'tC',0,'tD',tD,'
    sigTheta',0.00001,'sigB',0.0001);
6 sigM = 0.01; % Measurement noise
7
8 %% Design the controller
9 zeta = 1.0;
10 wN = 0.1;
11 kF = wN^2*dRHS.inertia;
12 tauR = 2*zeta*wN*dRHS.inertia/kF;
13 thetaOld = 0;
14
15 % Kalman Filter
16 dT = 1;
17 p = diag([0.001 0.0001]); % Initial covariance
18 q = 0.01*p; % Plant covariance
19 dKF = struct('p',p,'q',q,'r',sigM^2,...
20 'thetaE',0, 'xE',[0;0], 'a',[0 dT;0 0],...
21 'b',[dT;0], 'h',[1 0],...
22 'thetaGOld',0,'dT',dT);
23
24 % Time step
25 n = 1000;
26 xP = zeros(7,n);
27
28 %% Simulation
29 x = [0;0;0;0.1]; % [angle;rate;gyro angle;bias]
30 for k = 1:n
31     thetaM = x(1) + sigM*randn; % Measurement
32     thetaG = x(3); % Gyro angle
33     dKF = KalmanFilter(thetaG,thetaM,dKF);
34     dRHS.tC = -kF*(dKF.thetaE -
        thetaOld)/dT;
35     thetaOld = dKF.thetaE;
36     xP(:,k) = [x;dRHS.tC;dKF.thetaE;thetaM];
37     x = RK4(@RHS,x,dT,0,dRHS);
38 end
39
40 %% Plotting
41 t = (0:n-1)*dT;
42 yL = {'\theta','\omega','\theta_g','b','T_C',...
43 '\theta_e','\theta_m'};
44 TimeHistory(t,xP,yL,'SingleAxisSimulation')
45
46 %% Right hand side
47 function xDot = RHS(x,~,d)
48 omega = x(2);
49 b = x(4);
50 xDot = [omega;(d.tC+d.tD)/d.inertia;...
51 omega+b+d.sigTheta*randn;...
52 d.sigB*randn];
53 end
54
55 %% Kalman Filter
56 function d = KalmanFilter(thetaG,y,d)
57 % State propagation
58 omega = (thetaG - d.thetaGOld)/d.dT;
59 d.thetaGOld = thetaG;
60 d.xE = d.a*d.xE + d.b*omega;
61 d.p = d.a*d.p*d.a' + d.q;
62 % Measurement incorporation
63 k = d.p*d.h'/(d.h*d.p*d.h'+d.r);
64 d.xE = k*(y - d.h*d.xE);
65 d.p = (eye(2) - k*d.h)*d.p;
66 d.thetaE = d.xE(1);
67 end

```



Example 3.1: Single-axis step response.

3.7. Adding a mode

We can add a second wheel to the dynamical model. This could represent slosh or a mode due to structural flexibility. One way to derive the equations of motion of the new system is through angular-momentum conservation. Write the total angular momentum of the total spacecraft and that of just the second wheel.

$$H = I\dot{\theta} + J(\dot{\psi} + \dot{\theta}) \quad (3.18)$$

$$H_j = J(\dot{\psi} + \dot{\theta}) \quad (3.19)$$

where J is the inertia of the second wheel. The equations of motion are found by taking the derivatives of H and H_j .

$$T_d + T_c = I\ddot{\theta} + J(\ddot{\psi} + \ddot{\theta}) \quad (3.20)$$

$$T_j = J(\ddot{\psi} + \ddot{\theta}) \quad (3.21)$$

where T_j is the internal torque on the second wheel and ψ is the angle of the second wheel with respect to the core body. These can be written as

$$T_d + T_c = I\ddot{\theta} + T_j \quad (3.22)$$

$$T_j = J(\ddot{\psi} + \ddot{\theta}) \quad (3.23)$$

or

$$\begin{bmatrix} I+J & J \\ J & J \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} = \begin{bmatrix} T_c + T_d \\ T_j \end{bmatrix} \quad (3.24)$$

The matrix on the left side is known as the generalized inertia matrix. The generalized inertia matrix is symmetric. Let T_j be a general spring and damper torque,

$$T_j = -c\dot{\psi} - k\psi \quad (3.25)$$

We can then write out the state equations and get the eigenvalues. The eigenvalues are the natural frequencies of the system.

$$\begin{bmatrix} \dot{\theta} \\ \dot{\psi} \\ \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{k}{I} & 0 & \frac{c}{I} \\ 0 & -\frac{k}{J} & 0 & -\frac{c}{J} \end{bmatrix} \begin{bmatrix} \theta \\ \psi \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \quad (3.26)$$

where

$$\gamma = \frac{IJ}{I+J} \quad (3.27)$$

If $I \gg J$, $\gamma = J$. The script Example 3.2 shows the response with the added model. We compute the eigenvalues to determine the model frequency. The parameters, in this

```

1 %% Simulation of a single axis spacecraft
2 % Includes a wheel
3
4 %% Dynamical model data
5 tD = 1e-4; % Disturbance torque (Nm)
6 wJ = 0.6;
7 zetaJ = 0.1;
8 inertiaJ = 0.001;
9 k = inertiaJ*wJ^2;
10 c = 2*zetaJ*wJ*inertiaJ;
11 dRHS = struct('inertia',1,'tC',0,'tD',tD,'
    sigTheta',0.00001,...
12 'sigB',0.0001,'inertiaJ',inertiaJ,'k',k,'
    c',c);
13 sigM = 0.01; % Measurement noise
14
15 %% Design the controller
16 zeta = 1.0;
17 wN = 0.1;
18 kF = wN^2*dRHS.inertia;
19 tauR = 2*zeta*wN*dRHS.inertia/kF;
20 thetaOld = 0;
21
22 %% Eigenvalues
23 gamma = dRHS.inertiaJ*dRHS.inertia/(dRHS.inertiaJ
    + dRHS.inertia);
24 a = [0 0 1 0; 0 0 1;...
25 0 dRHS.k/dRHS.inertia 0 dRHS.c/dRHS.inertia;...
26 0 -dRHS.k/gamma 0 -dRHS.c/gamma];
27 eig(a)
28
29 % Kalman Filter
30 dT = 1;
31 p = diag([0.001 0.0001]); % Initial covariance
32 q = 0.01*p; % Plant covariance
33 dKF = struct('p',p,'q',q,'r',sigM^2,...
34 'thetaE',0,'xE',[0;0],'a',[0 dT;0 0]....
35 'b',[dT;0],'h',[1 0]....
36 'thetaOld',0,'dT',dT);
37
38 % Time step
39 n = 10000;
40 xP = zeros(9,n);
41
42 %% Simulation
43 x = [0;0;0;0;1;0;0]; % [angle;rate;gyro angle;
    bias]
44 for k = 1:n
45     thetaM = x(1) + sigM*randn; % Measurement
46     thetaG = x(3); % Gyro angle
47     dKF = KalmanFilter(thetaG,thetaM,dKF);
48     dRHS.tC = -kF*(dKF.thetaE + tauR*(dKF.thetaE -
        thetaOld)/dT);
49     thetaOld = dKF.thetaE;
50     xP(:,k) = [x;dRHS.tC;dKF.thetaE;thetaM];
51     x = RK4@dRHS,x,dT,0,dRHS);
52 end
53
54 %% Plotting
55 t = (0:n-1)*dT;
56 yL = {'\theta','\omega','\theta_g','b','T_C'...
57 '\theta_e','\theta_m','c','k'};
58 TimeHistory(t,xP,yL,'SingleAxisSimulation')
59
60 %% Right hand side
61 function xDot = RHS(x,-,d)
62 omega = x(2);
63 b = x(4);
64 psi = x(5);
65 psiDot = x(6);
66
67 tJ = -d.k*psi - d.c*psiDot;
68 omegaDot = (d.tC+d.tD - tJ)/d.inertia;
69 xDot = [omega;omegaDot;...
70 omega+b+d.sigTheta*randn;...
71 d.sigB*randn;...
72 psiDot;...
73 tJ/d.inertiaJ - omegaDot];
74 end
75
76 %% Kalman Filter
77 function d = KalmanFilter(thetaG,y,d)
78 % State propagation
79 omega = (thetaG - d.thetaGOld)/d.dT;
80 d.thetaGOld = thetaG;
81 d.xE = d.a*d.xE + d.b*omega;
82 d.p = d.a*d.p*d.a' + d.q;
83 % Measurement incorporation
84 k = d.p*d.h'/(d.h*d.p*d.h'+d.r);
85 d.xE = k*(y - d.h*d.xE);
86 d.p = (eye(2) - k*d.h)*d.p;
87 d.thetaE = d.xE(1);
88 end

```

```
1 >> SingleAxisWithWheelSim
```

```
2
```

```
3 ans =
```

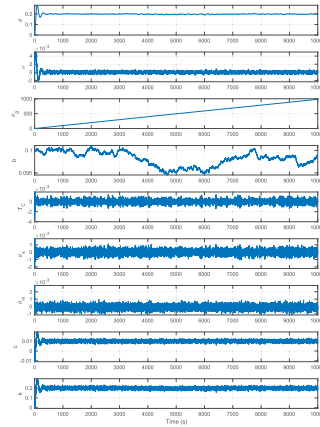
```
4
```

```
5 0.0000e+00 + 0.0000e+00i
```

```
6 0.0000e+00 + 0.0000e+00i
```

```
7 -6.0060e-02 + 5.9729e-01i
```

```
8 -6.0060e-02 - 5.9729e-01i
```



Example 3.2: Single-axis step response when there is a second mode. The eigenvalues are given. It has one oscillatory mode and one rigid-body mode.

example, do not represent any particular spacecraft. The undamped natural frequency of the mode is six times the controller bandwidth.

A step disturbance is applied to the spacecraft. The mode is excited but damps out quickly. The presence of the mode leads to a noisier response.

It is useful to look at the transfer function between torque and angle. Write

$$T = (I + J)s^2\theta + Js^2\psi \quad (3.28)$$

$$0 = J(s^2\psi + s^2\theta) + cs\psi + k\psi \quad (3.29)$$

where all angles are functions of s . Solving for ψ using the second equation

$$\psi = -\frac{Js^2\theta}{Js^2 + cs + k} \quad (3.30)$$

we get

$$T = (I + J)s^2\theta - \frac{Js^4\theta}{Js^2 + cs + k} \quad (3.31)$$

or

$$T = \left(\frac{(I + J)s^2(Js^2 + cs + k) - J^2s^4}{Js^2 + cs + k} \right) \theta \quad (3.32)$$

The angle response is therefore

$$\frac{\theta}{T} = \left(\frac{1}{(I + J)s^2} \right) \left(\frac{Js^2 + cs + k}{Js^2 + cs + k} \right) \quad (3.33)$$

The first part is the rigid-body response, which is a double integrator. The second part is a pole zero pair. If $I \gg J$ this is 1 since the zero will be very close to the pole. The zero will then cancel the pole, making it invisible to the control system. This analysis could be expanded to any number of modes. This particular example shows that the existence of a flex or slosh mode does not necessarily impact the closed-loop response. However, in every case, the influence needs to be determined through analysis.